

AI USAGE REPORT

AI-Assisted DBC Generation and CAN Data Visualization Using SocketCAN

1. AI Tool Selected

ChatGPT was used as an AI assistant during the development of the CAN and DBC project.

It was mainly used for:

- DBC generation and syntax
- CAN signal definition
- Python validation code
- SocketCAN transmitter development
- Troubleshooting
- DBC review

2. Prompts Used

The major prompts used during the project:

DBC Generation

Create a DBC file for a Vehicle Information Network containing Vehicle Speed, Engine RPM, Coolant Temperature, Fuel Level, and Battery Voltage. Define CAN IDs, signal lengths, scaling, offsets, ranges, and units.

DBC Validation

Create a Python script using cantools to validate the CAN messages and signal definitions in the DBC.

CAN Transmitter

Create a Python SocketCAN transmitter that generates changing vehicle values and sends them using the messages defined in the DBC.

DBC Review

Review the DBC for possible errors, signal layout issues, naming improvements, and missing documentation.

3. AI-Generated Outputs

AI assistance was used to generate or improve:

- Initial DBC message and signal definitions
- Python DBC validation script
- CAN decoding test script
- SocketCAN CAN transmitter
- Troubleshooting steps during implementation
- Suggestions for DBC improvements

The generated outputs were then tested on the actual Ubuntu system.

4. Corrections Performed

The AI-generated content was checked against the actual project results.

One issue was found during the Battery Voltage decoding test. The initial raw test data decoded to **12.58 V** while the expected value was **12.5 V**. The test data was corrected so that the expected and decoded values matched.

The DBC itself was also tested using **cantools**, and the final validation result was:

DBC VALIDATION: PASS

DECODING TEST: PASS

During the project, the DBC was also modified temporarily for Challenge 2 to observe the effect of an incorrect scaling factor. The original DBC was restored after the test.

5. Final Observations

Using AI reduced the time required to create the initial DBC and supporting Python code and was also useful for troubleshooting and understanding errors.

However, the project showed that AI-generated content cannot be accepted without verification. The DBC definitions, test results, CAN traffic, and decoded values were checked using **cantools**, SocketCAN, **candump**, and SavvyCAN.

The AI review also suggested possible improvements such as combining multiple signals into fewer CAN messages. This was not implemented because the current one-signal-per-message structure was simpler for this assignment.

Overall, AI was useful as an engineering assistant, while the final implementation and validation were performed independently.